

Supplementary Computational Provenance

Boundary-Localized Genomic Utility and Clinical-Tail Dominance

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Purpose

The main manuscript describes analyses in statistical rather than implementation-specific terms. This supplement records the computational provenance required to regenerate the figures and tables without introducing implementation filenames into the scientific narrative.

Figure-generation policy

All empirical manuscript figures are generated at compilation time with PGFPlots from tabular outputs in the repository analysis-results tree. No raster result figure is embedded in the manuscript. The figure source defines a single root path,

`../zebra_comp/RESULTS,`

so re-running the analyses and recompiling the manuscript updates plotted values without creating a separate manuscript-side data directory.

Theory notation used in the main text

The main theorem distinguishes two residual-genomic scales. $b_{\delta,n}$ is the finite-stage $(1 - \delta)$ control quantile of $|R_{G,n}|$ at a particular clinical-history stage n . The symbol b_δ is reserved for an n -independent bound that exists only under the stronger uniform-tightness condition A1-U. The empirical score-level analysis estimates an analogue of the former, not the latter.

Primary source mapping

Manuscript element	Analysis-output source under <code>zebra_comp/RESULTS</code>
Global repeated-split discrimination	<code>032_REPEATED_SPLITS_03_TARGET_LOGIC/auc_distribution_summary.csv</code> and <code>paired_auc_delta_summary.csv</code>
Paired incremental-value figure	<code>033_INCREMENTAL_LOGISTIC_32_COHORT/paired_incremental_summary.csv</code>
ZeBRA–MUC5B stratification	<code>MUC5B_STRATIFIED_BY_FILD/MUC5B_ZEBRA_STRATIFIED_SUMMARY.csv</code>
Local information figure	<code>LOCAL_ZEBRA_MUC5B_INFORMATION_CURVES/FPR_DEFINED_LOCAL_INFORMATION_BANDS.csv</code> and <code>LOCAL_ZEBRA_MUC5B_INFORMATION_WINDOWS.csv</code>
Boundary rescue figure	<code>MUC5B_ZEBRA_RESCUE_RULE/REPEATED_SPLIT_MUC5B_RESCUE_SUMMARY.csv</code> and <code>REPEATED_SPLIT_ADJUDICATED_MATCHED_FPR_SUMMARY.csv</code>
Stringent operating points	<code>032_REPEATED_SPLITS_03_TARGET_LOGIC/zedstat_operating_points_summary.csv</code>
Direct likelihood-ratio robustness figure	<code>EMPIRICAL_ZEBRA_MUC5B_LR_ROBUSTNESS/LR_ESTIMATOR_SENSITIVITY_GRID.csv</code> , <code>CLINICAL_TAIL_ROBUSTNESS.csv</code> , and <code>ROBUSTNESS_RANGE_SUMMARY.csv</code>

Table 1: Direct empirical likelihood-ratio summaries. The primary column uses the five-knot, $C = 1$ estimator; the range column reports the minimum and maximum across the nine prespecified spline and regularization specifications.

Quantity	Primary value	Robustness range
$b_{0.05}$	0.8699	0.8441–0.9083
$b_{0.01}$	0.9099	0.8635–0.9617
$b_{0.005}$	0.9241	0.8697–1.6100
Median $\log \Lambda_Z$ above 95th percentile	1.5175	1.4383–1.6069
95th pct. residual genomic scale above 95th percentile	0.9213	0.8155–5.6485
Fraction with $\log \Lambda_Z$ exceeding genomic scale above 95th percentile	0.9045	0.8811–1.0000
Median $\log \Lambda_Z$ above 97.5th percentile	2.7325	2.3756–2.8043
95th pct. residual genomic scale above 97.5th percentile	0.7758	0.7601–8.4206
Fraction with $\log \Lambda_Z$ exceeding genomic scale above 97.5th percentile	0.9938	0.8250–1.0000
Median $\log \Lambda_Z$ above 99th percentile	3.8345	3.7850–3.9379
95th pct. residual genomic scale above 99th percentile	0.7384	0.6849–0.7473
Fraction with $\log \Lambda_Z$ exceeding genomic scale above 99th percentile	1.0000	1.0000–1.0000

Direct empirical likelihood-ratio analysis

The score-level likelihood-ratio analysis is generated by

`test_empirical_zebra_muc5b_likelihood_ratios.py`

and the estimator-sensitivity analysis by

`test_empirical_zebra_muc5b_lr_robustness.py`.

Both use the same processed cohort inputs as the other analyses in `zebra_comp`. The primary direct-LR specification uses five spline knots and logistic regularization $C = 1$. The prespecified robustness grid crosses spline knots $\{4, 5, 6\}$ with $C \in \{0.3, 1, 3\}$; it is a sensitivity analysis rather than a model-selection exercise.

Direct likelihood-ratio robustness table

Theory–data scope

The empirical residual-genomic quantiles are fixed-score quantities for the observed ZeBRA representation and *MUC5B*. They correspond to the finite-stage $b_{\delta,n}$ construction in Theorem 1(ii), not to the optional uniform-tightness condition A1-U over all clinical-history lengths. Likewise, the observed high-ZeBRA likelihood-ratio tail does not establish the mathematical unboundedness assumption A2.

Exploratory local likelihood-ratio tests

The main manuscript reports the local nested-model likelihood-ratio statistics as descriptive localization summaries and omits nominal p-values because multiple regions and overlapping windows are examined in the same cohort. For reproducibility, the nominal one-degree-of-freedom p-values for the four non-overlapping control-FPR bands are: 2–5%: 0.0352; 5–10%: 4.74×10^{-5} ; 10–20%: 0.0289; and 20–40%: 2.15×10^{-4} . These values are not multiplicity-adjusted and are not used as confirmatory evidence.

Matched comparator within explicitly adjudicated negatives

The adjudicated-negative diagnostic reuses the saved outer-split seeds and the matched-ZeBRA thresholds learned from the original training data. The thresholds are applied unchanged to the explicitly adjudicated-negative subjects in the corresponding untouched test sets; the subset is never used for threshold fitting. The table below reports mean FPRs and rescue-minus-matched differences across 100 outer splits. The percentile range is descriptive across repeated splits rather than a confidence interval from an independent external sample.

Table 2: Explicitly adjudicated-negative comparison. FPR values are proportions; Δ is rescue minus matched ZeBRA.

Rule	Baseline FPR	Rescue FPR	Matched FPR	Mean Δ	2.5–97.5% Δ
0.5% / 2%	0.1292	0.1421	0.1458	-0.0037	-0.0249–0.0094
0.5% / 5%	0.1292	0.1628	0.1689	-0.0060	-0.0353–0.0225
0.5% / 10%	0.1292	0.1958	0.2144	-0.0185	-0.0561–0.0194
1% / 2%	0.1548	0.1635	0.1628	+0.0006	-0.0142–0.0138
1% / 5%	0.1548	0.1842	0.1888	-0.0046	-0.0315–0.0194
1% / 10%	0.1548	0.2172	0.2364	-0.0192	-0.0523–0.0188

For the 0.5%/10% rule, rescue has lower adjudicated-negative FPR than matched ZeBRA in 81% of outer splits; the corresponding fraction is 78% for the 1%/10% rule. Five of the six rules have a negative mean rescue-minus-matched FPR, while the 1%/2% rule is essentially equal. Because the adjudicated-negative subset contains only about 216 subjects per outer split on average, these results resolve the earlier concern that rescue might uniquely inflate adjudicated-negative FPR but do not establish a statistically definitive reduction.

Calibration summary

The raw ZeBRA score is not an absolute calibrated FILD/FILA probability in this cohort. Across repeated held-out splits, the mean calibration slope is 0.180 for ZeBRA, 0.241 for the genomic model, and 1.140 for the combined model; corresponding expected calibration errors are 0.570, 0.115, and 0.121. The main manuscript therefore treats ZeBRA as an ordinal clinical-history score unless an explicit score-level likelihood-ratio transformation is estimated.

Supporting ROC-hull summary

The descriptive ROC-hull analysis is retained as a supporting result rather than a primary predictive comparison. The empirical/interpolated AUCs are 0.8142 for ZeBRA and 0.8095 for the hybrid model; the convex hulls of the individual curves have AUCs 0.8219 and 0.8182, respectively; the pointwise maximum has AUC 0.8196; and the convex hull of the concatenated ZeBRA and hybrid ROC points has AUC 0.8260. These hull quantities are geometric envelopes over the evaluated ROC point sets and are not held-out performance estimates from a separately fitted classifier. The concatenated hull is included to illustrate the best randomized operating trade-off available from the observed ZeBRA and hybrid points; it is neither a universal theoretical maximum nor evidence that a classifier with AUC 0.8260 has been prospectively validated.

Authoritative manuscript sources

The canonical manuscript master is `zebra_genomics_boundary_theorem_empirical.tex`. The scientific narrative is split across `manuscript_body_intro.tex`, `manuscript_body_theory.tex`, `manuscript_body_interpretation.tex`, `manuscript_body_empirical_a.tex`, `manuscript_body_empirical_b.tex`, and `manuscript_body_end.tex`. Figure definitions are assembled by `empirical_figures.tex` from `empirical_figures_data.tex`, `empirical_figures_primary.tex`, `empirical_figures_lr.tex`, and `empirical_figures_adjudicated.tex`. These figure-source files are the authoritative mapping between manuscript plots and analysis-result tables.